ChronoFork⏳🔀:

Learner–Multi-Agent Co-Roleplay for Divergent Historical Experiences

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IS492 - Spring 2026
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https://github.com/IS492-SP26/team-project-chronofork-history-rewrite-simulation

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1. Recap & Final Goals

👥 Who we're building for

- History learners
- Engaging counterfactual reasoning exp
- Not formal classrooms

❌ ChatGPT / Claude — great at subtasks

- But can't create a learning EXPERIENCE

MOST REWARDING FEEDBACK

"It turned me from a reader into an author."

— P4, post-study interview

BACKED BY THE NUMBERS

+3.11 on Sense of Agency (PENS, 1–7)

6 / 6 participants moved in the same direction

2. Evaluation Design Overview

Participants

- N = 6 (P1–P6, university students 18–25)
- Within-subject, counter-balanced order
- Each chose 2 events from the Event Library; 1 per system
- Compensation: $10 / hr
- Recruited via online groups

Two Conditions

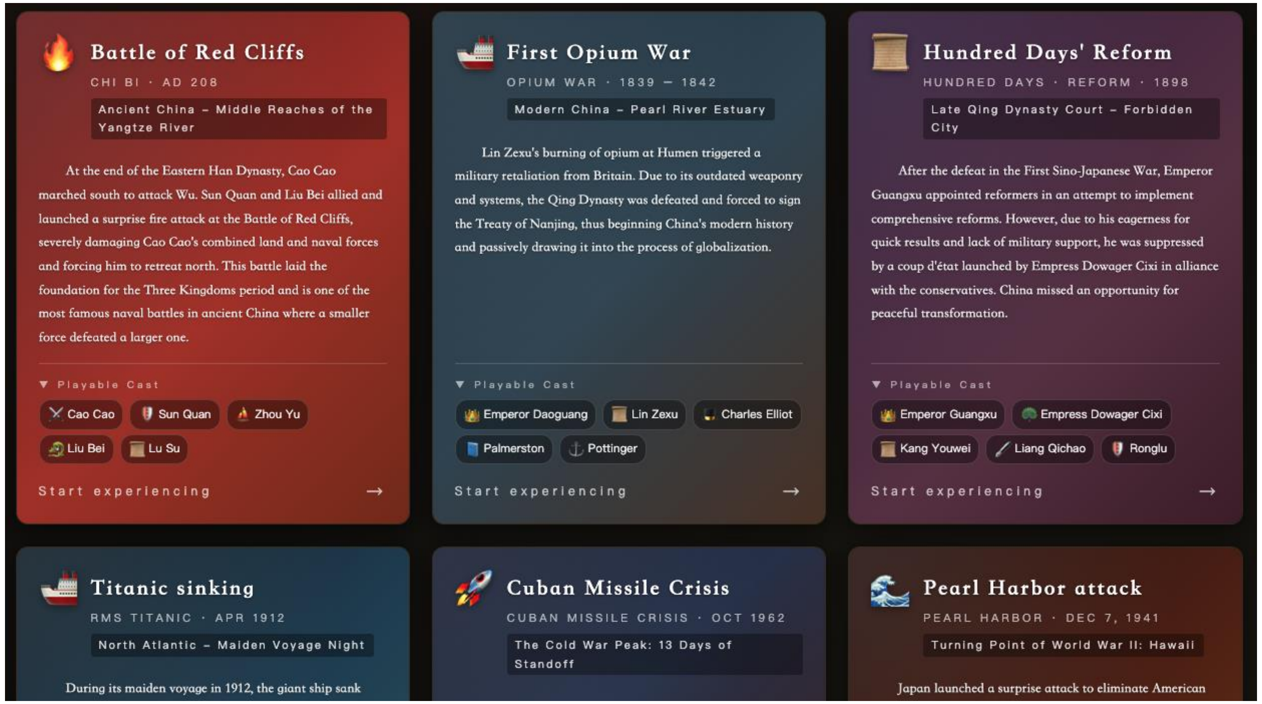
- ChronoFork-Web (dynamic)
- IDN-Twine (static)
- Both cano expert-audit
- Lab sessio

Metrics Collected

- NES (12 items) — Narrative Engagement
- ESS (7 items) — Explanation Satisfaction
- PENS Autonomy (3 + 2 items) — Agency
- UMUX-Lite (2) + Paas Mental Effort (1)
- IMI — Interest, Value, Pressure
- Semi-structured interview (~20 min)

3. Study Materials & Procedure

1. Intro & Consent	2. Task — Exp. A	3. Survey + Break	4. Task B + Q2 + Interview
~5 min briefing, sign consent, event selection	~17 min tutorial video → 15 min think-aloud	~7 min Q1 (~5 min) + 2-min rest	~30 min mirror Exp. A, then ~20 min interview



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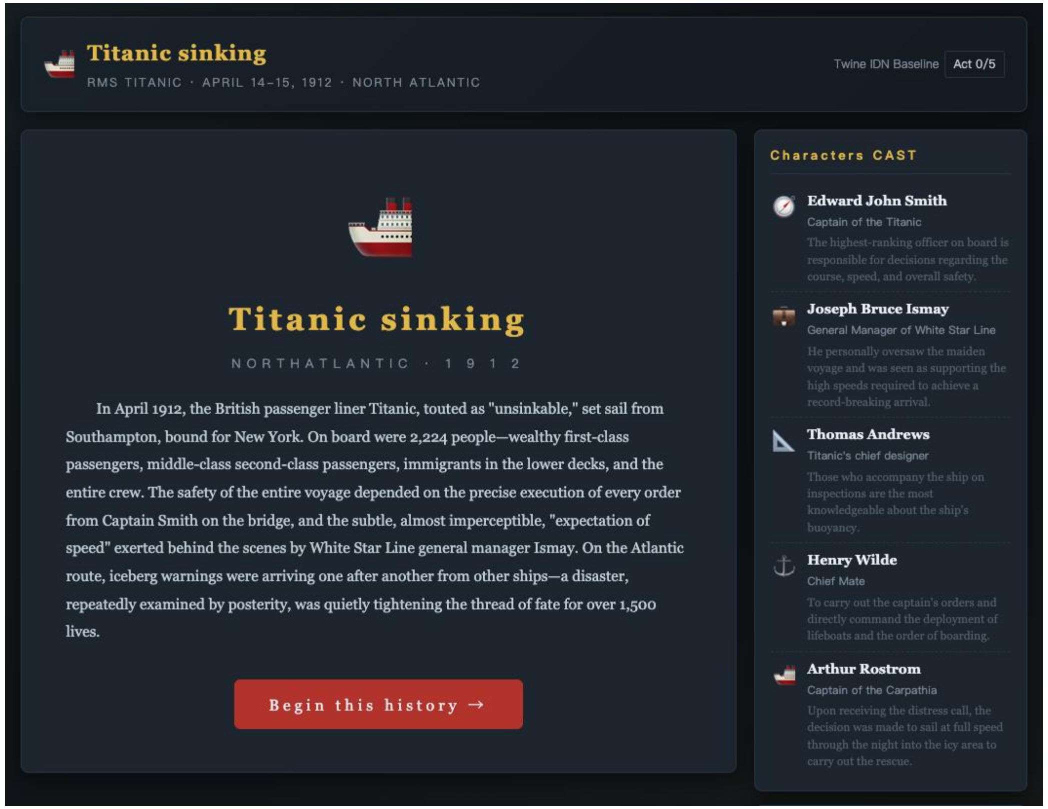
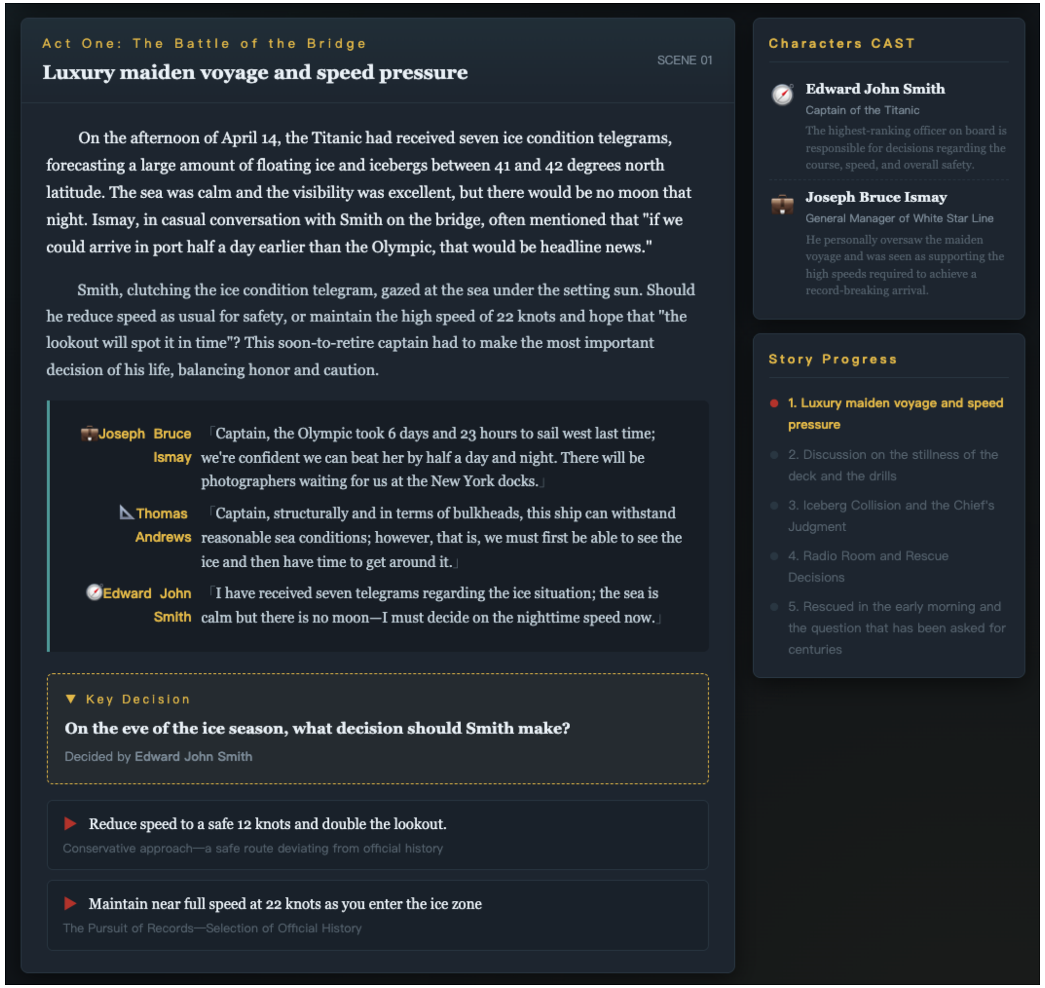


Baseline (IDN)

(twine.chronofork.me)



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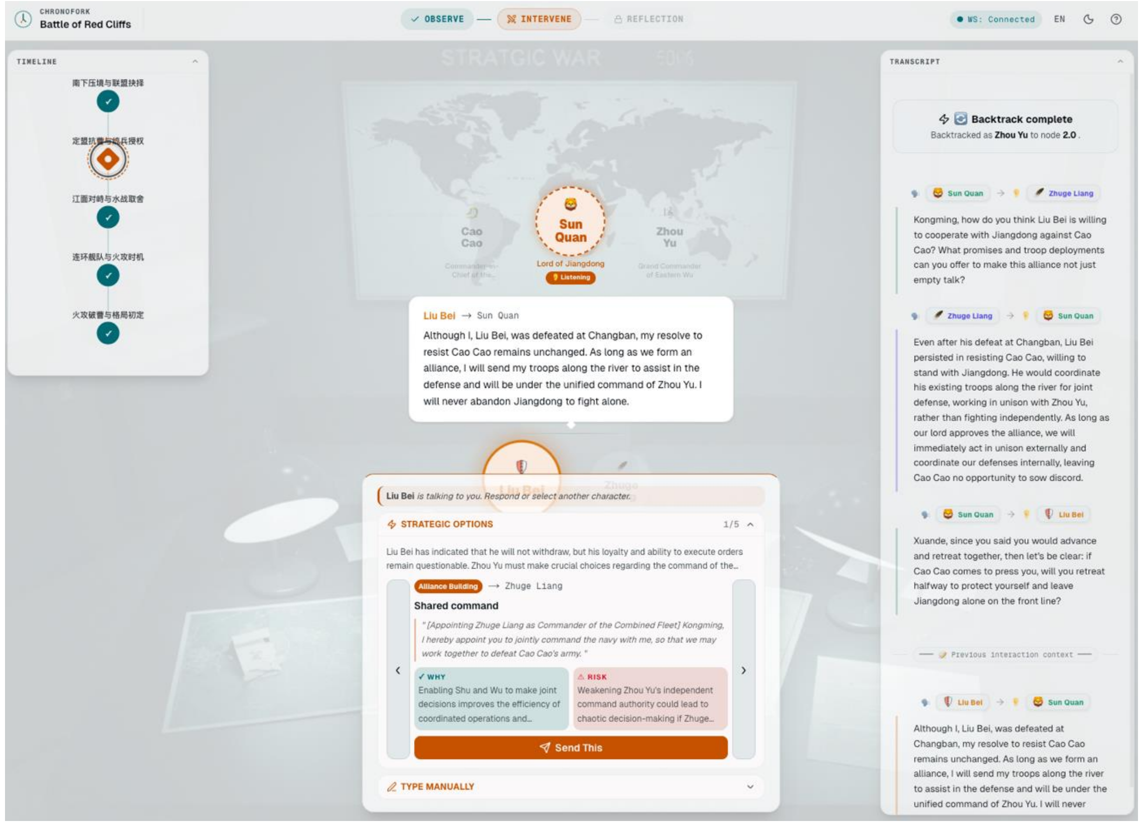
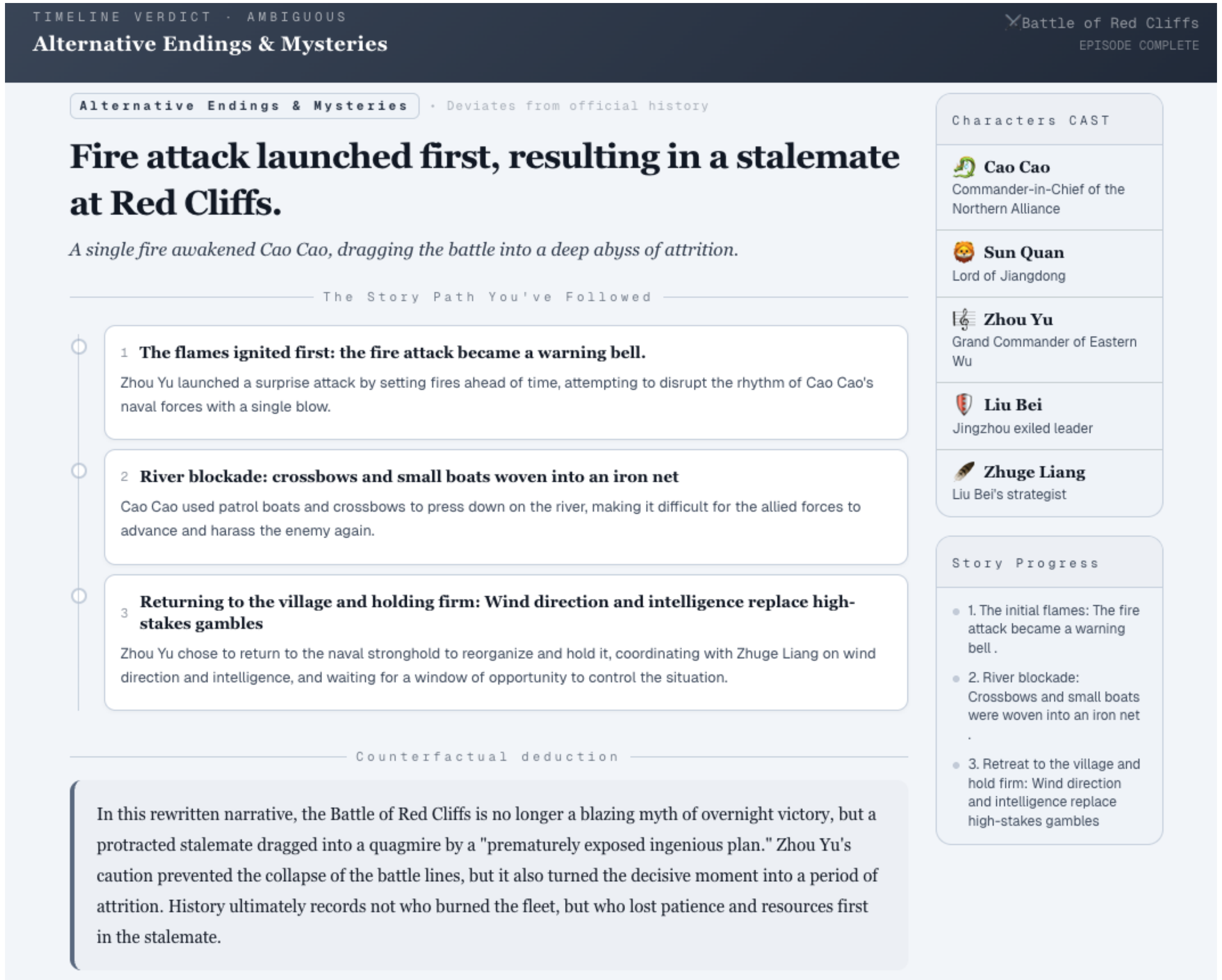
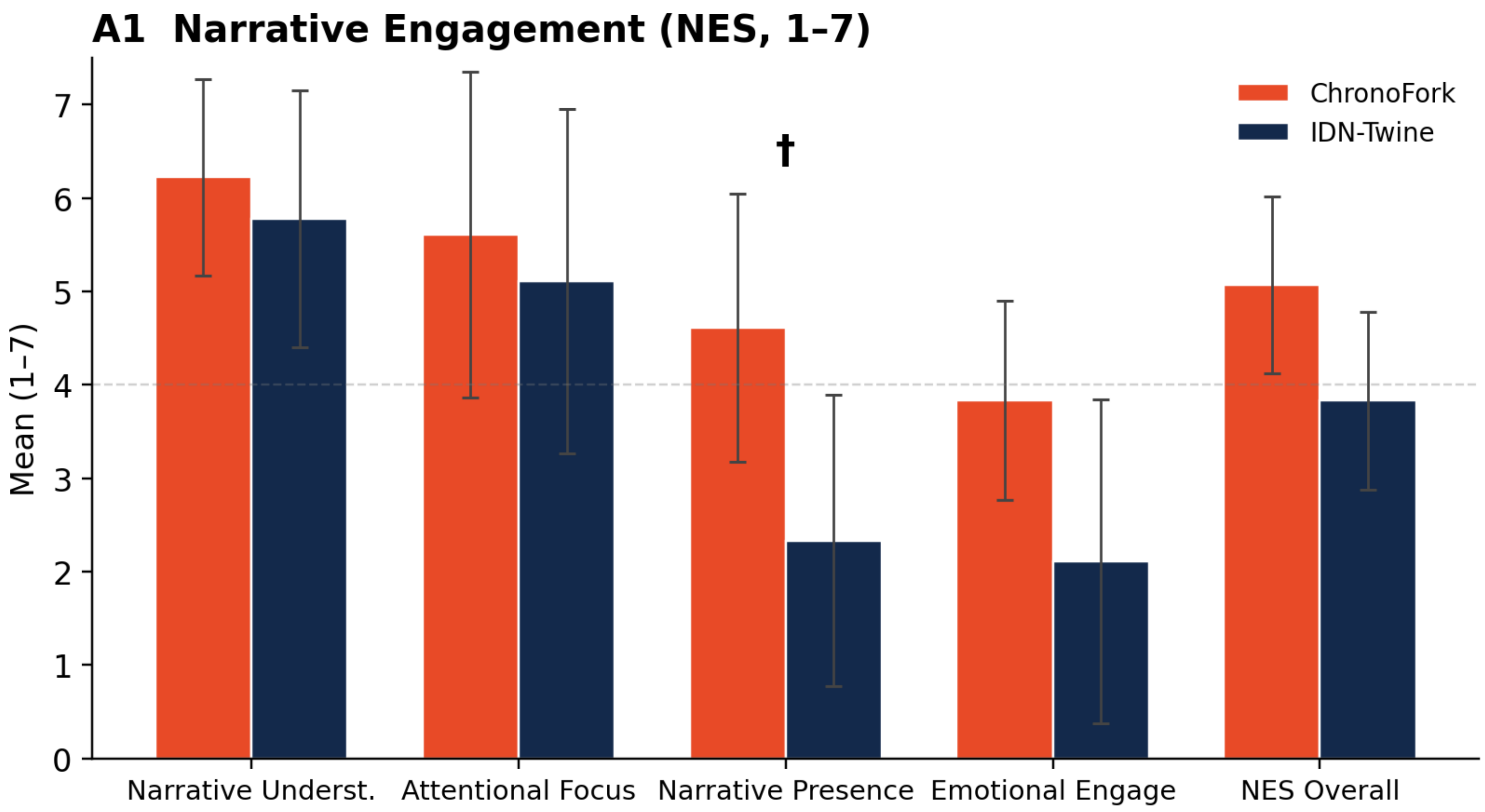


ChronoFork Web
(app.chronofork.me)



4. Quantitative Results

Narrative Engagement (N = 6, Wilcoxon two-sided)



Narrative Presence +2.28

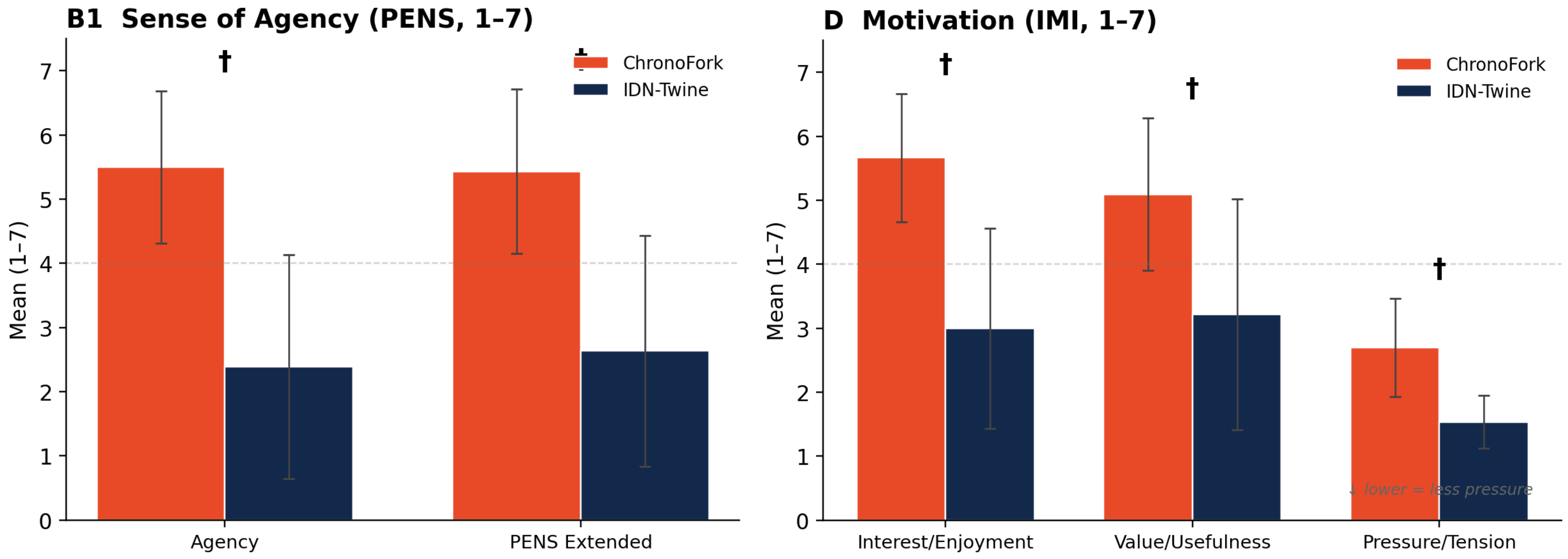
p = .094

† p < .10

N=6 → smallest possible p (two-sided) is .063

4. Quantitative Results

Sense of Agency (N = 6, Wilcoxon two-sided)



Agency +3.11

p = .063 (6 / 6 same direction)

Interest / Enjoyment +2.67

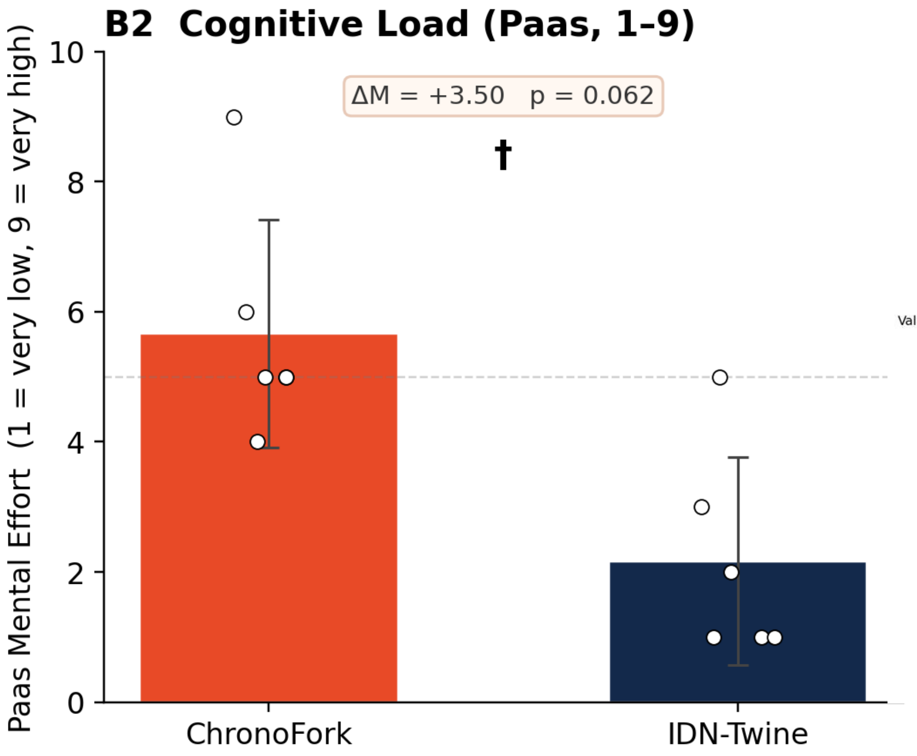
p = .063

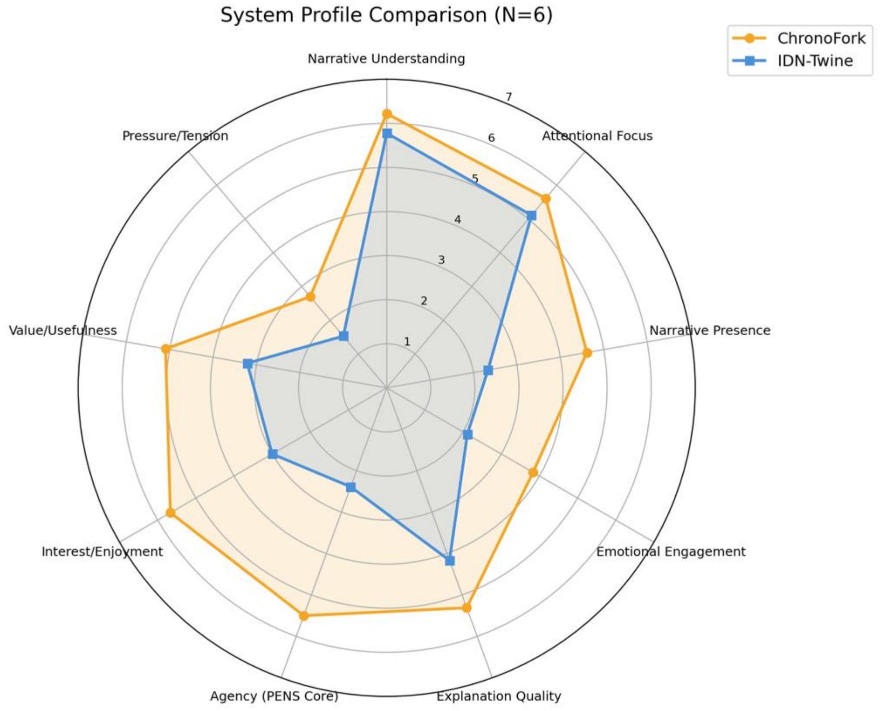
† p < .10

N=6 → smallest possible p (two-sided) is .063

4. Quantitative Results

Sense of Agency (N = 6, Wilcoxon two-sided)





Mental Effort +3.50 (1–9)

p = .063 higher = more effort

UMUX-Lite −0.08

(66.7 vs 68.1 / 100)

p = 1.00 honest tie

† $p < .10$

N=6 → smallest possible p (two-sided) is .063

5. Interpretation & Discussion

Orchestration delivered the experience

- Agency, Presence, Interest, Value — all 6 / 6 same direction
- Effect sizes large ($r \geq 0.77$) on every key narrative dimension
- "Reader → author" reframing reported by all 6 interviewees
- CP3 success criteria (engagement + critical thinking) supported relative to IDN baseline
- Confirms CP2's thesis: the missing layer was orchestration, not generation.

Higher mental effort may = germane, not extraneous

- Paas mental effort: ChronoFork 5.67 vs. Twine 2.17
- But UMUX-Lite is a tie → the load is not from the UI
- Pressure / Tension also slightly higher → active improvisation
- P2: "I had to think on my feet — that's why it felt alive."
- Reframe: the cost of being an author is doing the authoring.

6. Limitations, Risks & Ethics

Sample

- N = 6 pilot scale only
- Mandarin-speaking university students 18–25
- Convenience sample — campus posters & online groups
- Counter-balanced order, but practice effects possible

Methodology

- Open-ended exploration → no strict task-success metric
- 15-min cap per condition → may favor quick-to-read systems
- Lab setting ≠ real-world self-paced learning

Model & System Risk

- Persona drift observed in 4 / 6 (Khrushchev, Cao Cao, Ronglu, Titanic crew)
- 5–15 s response latency disrupted attention
- Reproducibility: cached LLM responses + 25 unit tests

Ethics & Privacy

- Informed consent obtained (logs, audio, screen)
- PII removed during transcription
- Compensation: $10 / hr
- Conversations sent to OpenAI API only (no third-party storage)

7. Conclusion & Future Work

TAKEAWAY 1

Orchestration beats better generation.

CP2 already showed LLMs nail subtasks. CP4 shows the missing layer was a learner-in-the-loop workflow — and once it's there, agency and engagement separate sharply from a static IDN baseline.

TAKEAWAY 2

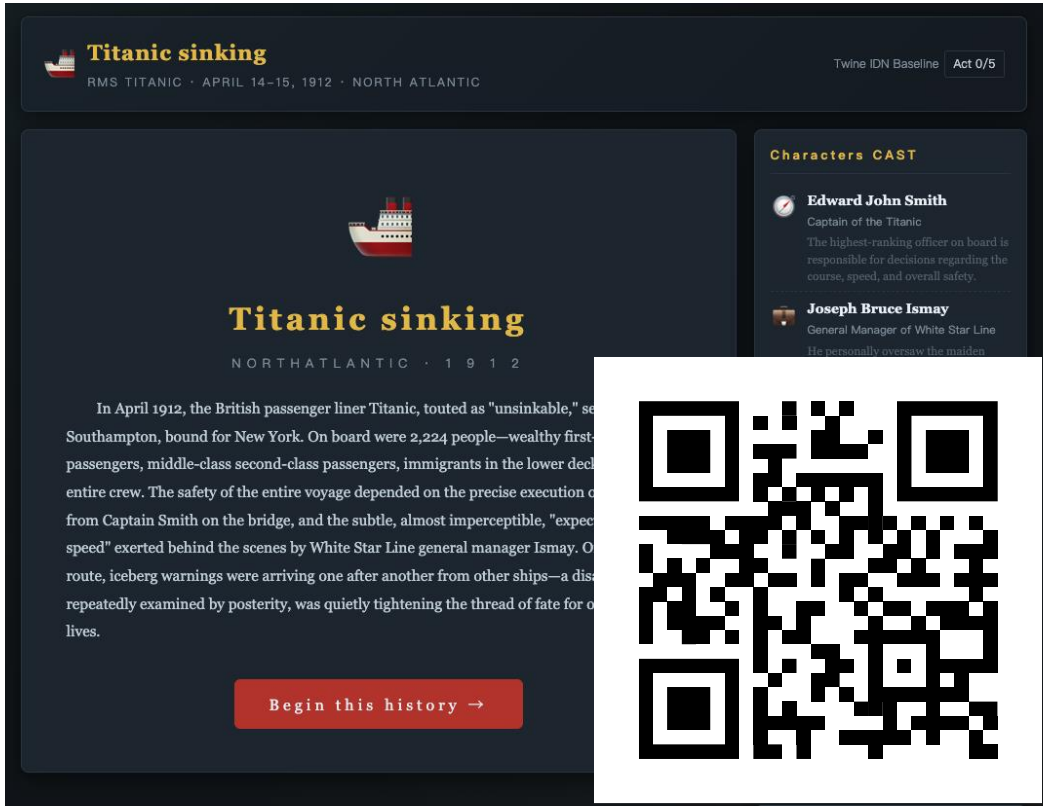
Reading is the bottleneck — visualization is the lever.

Text fatigue was the most consistent friction across all six participants. Multi-voice dialogue is what makes the orchestration feel alive, but it's also why people get tired. The clearest leap forward is visual — portraits, scenes, consequence panels — and ultimately embodied XR, where dialogue happens in space rather than on a wall of words.

FUTURE WORK

- **Persona stability layer**
  Skill-tagged actors (P3); knowledge-state grounding; persona-break detection
- **Reflection / outcome panel**
  Explicit canonical-vs-counterfactual diff (P3, P6)
- **Latency & onboarding**
  Streaming, "where am I" cues, 30-s walkthrough
- **Larger evaluation**
  20+ participants; retention test 1 week post-study
- **Immersive thread**
  Quest 3 prototype reactivated once web is stable

QA & Try out



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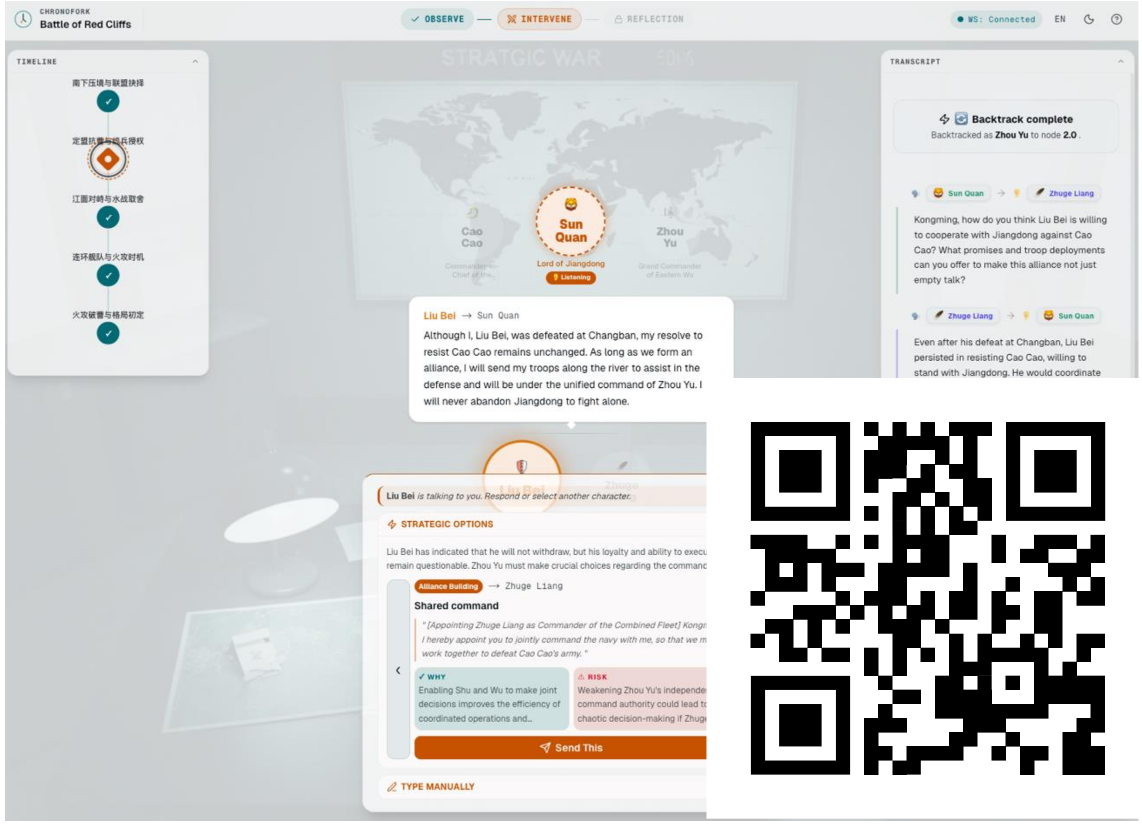

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